StudySync — Peer Matching & Collaborative Study Group Platform

Project Management: Team Meeting Log & Weekly Progress Record

Course: Software Engineering & Project Management

Project Group: Group 15 (Section F)

Target System: StudySync Web Platform

Period Covered: Weeks 1 through 6

Attendance Policy: Mandatory Weekly Scrum Coordination with 100% Member Quorum

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1. Meeting Governance & Recordkeeping Policy

In accordance with Section 3.E of the Course Evaluation Guidelines, all project engineering meetings are conducted following Agile Scrum conventions:

- **Frequency:** Weekly physical or hybrid sprint planning, review, and retrospective sessions supplemented by daily asynchronous coordination.
- **Quorum:** All four team members must attend and participate actively to ensure equal contribution distribution.
- **Documentation Standard:** Each meeting record captures the meeting timestamp, modality, roll call, agenda items, key technical debates resolved, formal decisions adopted, and individual deliverable assignments.

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2. Attendance & Participation Summary Matrix

Meeting ID	Project Week & Date	Session Focus / Milestone	Modality	Member 1 (Frontend)	Member 2 (Backend)	Member 3 (Database)	Member 4 (QA/Lead)	Quorum Status
M-01	Week 1 (Sep 05, 2026)	Kickoff, Problem Definition, Tech Stack	In-Person	Present	Present	Present	Present	**100% Quorum**
M-02	Week 2 (Sep 12, 2026)	IEEE SRS, 3NF Schema & API Specification	Hybrid	Present	Present	Present	Present	**100% Quorum**
M-03	Week 3 (Sep 19, 2026)	Sprint 1 Review & Increment 1 Sign-Off	In-Person	Present	Present	Present	Present	**100% Quorum**
M-04	Week 4 (Sep 26, 2026)	Sprint 2 Planning & Matching Engine Math	Hybrid	Present	Present	Present	Present	**100% Quorum**
M-05	Week 5 (Oct 03, 2026)	Concurrency Row-Locking & Integration Check	In-Person	Present	Present	Present	Present	**100% Quorum**
M-06	Week 6 (Oct 16, 2026)	Mid-Sem Evaluation Audit & Demo Sign-Off	In-Person	Present	Present	Present	Present	**100% Quorum**

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3. Chronological Meeting Records (Weeks 1 to 6)

Meeting 01 (Week 1): Project Kickoff, Scope & Architecture Selection

- **Date & Time:** September 05, 2026 | 14:00 – 15:45 (105 mins)
- **Location:** Hostel Room
- **Attendees:** Member 1, Member 2, Member 3, Member 4 (All Present)
- **Agenda:**
  1. Detailed analysis of course project guidelines and mid-semester evaluation criteria (40-60% target).
  2. Brainstorming student academic pain points regarding peer discovery and study group formation.
  3. Definition of core in-scope features vs. explicit out-of-scope features (e.g., excluding in-browser WebRTC video).
  4. Technology stack evaluation: React 18 + Vite + Tailwind CSS, Node.js + Express, PostgreSQL 14+.
- **Key Technical Decisions Adopted:**
  - **Dec-01.1:** Adopted Three-Tier Layered Architecture (Client SPA → Express REST Gateway → PostgreSQL 3NF).
  - **Dec-01.2:** Established mandatory four-member task partitioning (25% planned effort per member).
  - **Dec-01.3:** Enforced strict git branching convention (`feature/<module>-<description>`) with peer pull request reviews.

- **Action Items & Commitments:**
  - *Member 1:* Setup Vite React 18 boilerplate and configure Tailwind CSS tokens.
  - *Member 2:* Initialize Express server directory structure and install runtime middleware (CORS, dotenv).
  - *Member 3:* Draft entity relationships and define preliminary database entities.
  - *Member 4:* Draft initial IEEE Std 830-1998 SRS outline and setup master traceability matrix.

Meeting 02 (Week 2): SRS Formalization, 3NF PostgreSQL DDL & API Contracts

- **Date & Time:** September 12, 2026 | 15:00 - 16:30 (90 mins)
- **Location:** Hybrid
- **Attendees:** Member 1, Member 2, Member 3, Member 4 (All Present)
- **Agenda:**
  1. Review of IEEE Std 830-1998 SRS functional requirements (`FR-001` through `FR-020`).
  2. Review of Third Normal Form (3NF) relational schema and UUID primary keys.
  3. Formalization of API contract specification and standardized JSON response envelope.
  4. Design of weekly recurring availability representation in PostgreSQL.
- **Key Technical Decisions Adopted:**
  - **Dec-02.1:** Standardized on uniform JSON response envelope: `{ success: boolean, data: any, error: object, meta: object }`.
  - **Dec-02.2:** Decided to store student weekly availability as structured `JSONB` in `user_profiles` with Generalized Inverted Index (GIN) rather than a rigid minute-level relational table.
  - **Dec-02.3:** Enforced bcrypt password hashing with minimum 10 work rounds prior to database persistence.
- **Action Items & Commitments:**
  - *Member 1:* Create TypeScript interfaces in `frontend/src/types/index.ts` matching API specifications.
  - *Member 2:* Implement Zod validation schemas for registration, login, and profile update endpoints.
  - *Member 3:* Author baseline SQL migration script `001_initial_schema.sql` and seed catalog data.
  - *Member 4:* Define Master Test Case Register (`TC-001` to `TC-020`) and setup Jest/Supertest environment.

Meeting 03 (Week 3): Sprint 1 Review — Auth, Profiles & Group Catalog

- **Date & Time:** September 19, 2026 | 14:00 - 15:30 (90 mins)
- **Location:** Hostel Room
- **Attendees:** Member 1, Member 2, Member 3, Member 4 (All Present)
- **Agenda:**
  1. Demonstration of working Increment 1 features: User Registration, JWT Login, Profile Setup, and Group Directory.
  2. Integration testing review: Verification of Supertest integration tests for `/auth` and `/groups`.
  3. Retrospective on initial frontend-backend contract synchronization.
- **Key Technical Decisions Adopted:**
  - **Dec-03.1:** Accepted Increment 1 into `develop` branch baseline.
  - **Dec-03.2:** Added client-side Axios response interceptors to automatically unwrap response data envelopes and manage 401 unauthorization redirects.
- **Action Items & Commitments:**
  - *Member 1:* Develop `JoinRequestModal` and `OwnerReviewPanel` React views for Sprint 2.
  - *Member 2:* Implement membership routes (`POST /join`, `GET /requests`, `PATCH /members/:userId`).
  - *Member 3:* Author 3-factor heuristic matching algorithm in `matching.service.js`.
  - *Member 4:* Write automated unit test suites for compatibility math (`tests/matching.test.js`).

Meeting 04 (Week 4): Sprint 2 Kickoff — Membership Workflow & Heuristic Math

- **Date & Time:** September 26, 2026 | 15:30 - 17:00 (90 mins)
- **Location:** Hostel Room
- **Attendees:** Member 1, Member 2, Member 3, Member 4 (All Present)
- **Agenda:**
  1. Detailed breakdown of user stories `US-05` (Join Request), `US-06` (Owner Review), and `US-07` (Compatibility Recommendations).
  2. Mathematical formulation of the 3-Factor Compatibility Score:

$$\text{Score} = (S_{\text{subject}} \times 0.40) + (S_{\text{schedule}} \times 0.40) + (S_{\text{goals}} \times 0.20)$$

  3. Strategy for handling group capacity boundaries and preventing over-enrollment.

- **Key Technical Decisions Adopted:**
  - **Dec-04.1:** The matching engine must return an explainable criteria breakdown `{ subjectScore, scheduleScore, goalsScore, totalScore }` to build user trust.
  - **Dec-04.2:** Duplicate join requests from the same user to the same group must be rejected with HTTP 409 Conflict.
- **Action Items & Commitments:**
  - *Member 1:* Build dynamic character counter on applicant motivation notes (max 500 chars).
  - *Member 2:* Author authorization middleware verifying requesting user is the true group creator.
  - *Member 3:* Implement slot interval intersection logic in `matching.service.js`.
  - *Member 4:* Author automated integration test suite for membership endpoints (`tests/membership.test.js`).

Meeting 05 (Week 5): Mid-Sprint Integration — Concurrency & Capacity Locking

- **Date & Time:** October 03, 2026 | 14:00 - 16:00 (120 mins)
- **Location:** Hostel Room
- **Attendees:** Member 1, Member 2, Member 3, Member 4 (All Present)
- **Agenda:**
  1. End-to-end integration test of membership workflow in local browser environment.
  2. Analysis of potential race conditions when groups near maximum member capacity ($n = \text{max_members}$).
  3. Resolution of database transaction boundaries for member approvals.
- **Key Technical Decisions Adopted:**
  - **Dec-05.1:** Implemented transactional row-level locking (`SELECT ... FOR UPDATE`) in `GroupRepository.updateMembershipStatus` to eliminate race conditions during concurrent approval clicks.
  - **Dec-05.2:** Mandated 100% test pass rate gate before proceeding to mid-semester documentation compilation.
- **Action Items & Commitments:**
  - *Member 1:* Finalize responsive mobile layout verification for review modal and directory views.
  - *Member 2:* Optimize centralized error envelope formatting for concurrency conflict errors.
  - *Member 3:* Execute database query explain plans verifying sub-25ms index scan latencies.
  - *Member 4:* Execute comprehensive 43-test regression run and capture raw CLI logs in `docs/testing/evidence/`.

Meeting 06 (Week 6): Sprint 2 Review & Mid-Semester Audit (50% Milestone)

- **Date & Time:** October 16, 2026 | 16:30 - 18:30 (120 mins)
- **Location:** Hostel Room
- **Attendees:** Member 1, Member 2, Member 3, Member 4 (All Present)
- **Agenda:**
  1. Full mid-semester progress audit against Course Evaluation PDF criteria (Page 1 & 2).
  2. Verification of working end-to-end feature: Registration → Availability → Search → Join Request → Owner Approval.
  3. Review of test execution results: 43 automated tests passed (100% pass rate, 0 failures).
  4. Final assembly and sign-off on submission documents (Report, SRS, Diagrams, Test Cases, Plan, Video Link).
- **Key Technical Decisions Adopted:**
  - **Dec-06.1:** Confirmed **50% Project Completion Milestone Achieved** (Increment 1 & 2 operational).
  - **Dec-06.2:** Approved final demonstration video script (5–7 minutes duration) and verified cloud view-only permissions.
  - **Dec-06.3:** Verified individual contribution declaration signed by all 4 members at 25% equal effort.
- **Action Items & Commitments for Sprint 3 Kickoff:**
  - *Member 1:* Design calendar view wireframes with Month/Week/Day filters.
  - *Member 2:* Scaffold `study_sessions` API endpoints and RSVP route handlers.
  - *Member 3:* Design calendar collision detection interval query.
  - *Member 4:* Setup Sprint 3 test suites and assemble final ZIP submission package.

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4. Remaining Semester Meeting Schedule (Weeks 7 to 12)

Planned Meeting	Target Week	Primary Agenda & Technical Focus	Expected Deliverable
Meeting 07	Week 7 (Oct 24, 2026)	Sprint 3 Kickoff: Study Session Scheduling & Venue Validation	Session DDL and API routes
Meeting 08	Week 8 (Oct 31, 2026)	Calendar Conflict Detection Algorithm & Overlap Warnings	Collision detection test suite

Planned Meeting	Target Week	Primary Agenda & Technical Focus	Expected Deliverable
Meeting 09	Week 9 (Nov 07, 2026)	Sprint 3 Review: Interactive Calendar & Resource Sharing	Full Increment 3 operational baseline
Meeting 10	Week 10 (Nov 14, 2026)	Sprint 4 Kickoff: Full System E2E Regression Testing	Master regression execution log
Meeting 11	Week 11 (Nov 21, 2026)	Security Vulnerability Audit (OWASP Top 10) & Dockerizing	Docker Compose bundle & audit report
Meeting 12	Week 12 (Nov 28, 2026)	Final Project Packaging, User Manual & Master Demo	Final 100% Release Package

5. Team Sign-Off & Verification Declaration

We, the undersigned team members of **StudySync (Group 15, Section F)**, hereby certify that:

1. The meeting logs documented above accurately reflect the discussions, decisions, and weekly progression of our team throughout Weeks 1 to 6.
2. All team members attended all scheduled meetings with 100% quorum.
3. Every member has independently authored and verified their assigned technical modules.

Member Name	Student Registration No.	Assigned Primary Role	Attendance Rate	Confirmation Signature
Yadavalli Sumanth Chowdary	`20243322`	Frontend Engineering & UI/UX	100% (6/6)	*Confirmed (2026-10-16)*
Vuppalapanchu Karthikeya Lalith	`20243321`	Backend API & Authentication	100% (6/6)	*Confirmed (2026-10-16)*
Vem Nagarjuna Reddy	`20243315`	Database & Matching Engine	100% (6/6)	*Confirmed (2026-10-16)*
Tummala Ashrith Reddy	`20243297`	Scheduling, QA & Lead	100% (6/6)	*Confirmed (2026-10-16)*